VOICE CONTROLLED HOME AUTOMATION USING ARDUINO AND BLUETOOTH MODULE

FINAL REPORT

Submitted for the course

By

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ACKNOWLEDGEMENT

We would like to express our gratitude to all those who provided us the possibility to complete this project. A special gratitude I give to the faculty of this course, whose contribution in stimulating suggestions and encouragement, helped us to coordinate our project especially in doing this project. We have to appreciate the guidance given by him especially in our project presentation that has improved our presentation skills thanks to his comment and advices.

OBJECTIVE

Our main objective is to how to control an AC light and DC fan on Proteus software through voice commands on mobile application

It is divided into following sections

Circuit designing in Proteus

Controlling circuit through app on Android phone.

ABSTRACT

- Home automation is one of the major growing industries that can change the way people live.
- Some of these home automation systems target those seeking luxury and sophisticated home automation platforms; others target those with special needs like the elderly and the disabled.
- Typical wireless home automation system allows one to control house hold appliances from a centralized control unit which is wireless.

LITERATURE REVIEW:

In Bluetooth based home automation system the home appliances are connected to the Arduino BT board at input output ports using relay. The connection is made via Bluetooth. The Bluetooth connection is established between Arduino BT board and phone for wireless communication. According to Muhammad Asadullah and Khalil Ullah (2017), "In proposed work ultrasonic sensor is used for the measurement of water level inside the water tank. Soil moisture sensor is used to measure the water content inside the soil"(p.3). We have not used the ultrasonic sensor and the Soil moisture sensor as mentioned in the paper due to the ongoing situation and lack of resources. We have made a simple system which controls the fan and light of a household. The proposed method presents the design and implementation of a robust, low cost and userfriendly home automation system using Bluetooth technology. It also describes the hardware and software architecture of system, future work and scope. The proposed prototype of home automation system can be implemented and tested on hardware and it can gave the exact and expected results. A Bluetooth based wireless home automation system can be implement with a low cost and it is easy to install in an existing home. A research work proved that Bluetooth system are faster than wireless and GSM systems.

COMPONENTS AND SOFTWARE

SOFTWARE : PROTEUS

1) Arduino Uno
2) Bluetooth module
3) Lamp
4) Motor
5) AC Supply
6) Relay
7) Resistors

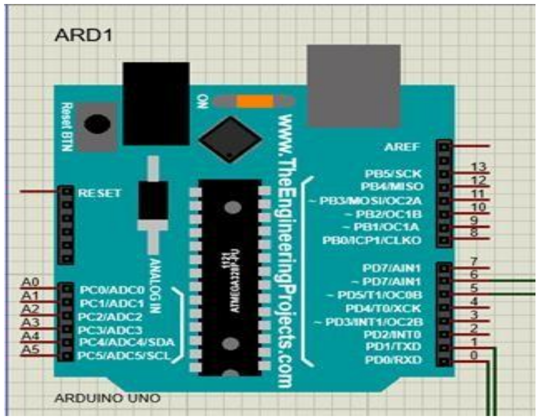
COMPONENTS DESCRIPTION:

Arduino UNO

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits.

The applications of Arduino Uno include the following. Arduino Uno is used in Do-it-Yourself projects prototyping. In developing projects based on code-based control. Development of Automation System. Designing of basic circuit designs.

- Arduino is used to power the relay modules.
- Arduino is programmed to receive instructions from bluetooth module.
- Arduino powers the relay modules which are connected to pins 5 and 6 based on the instructions given by the Bluetooth module.



Arduino IDE

The Arduino Integrated Development Environment contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload programs and communicate with them.

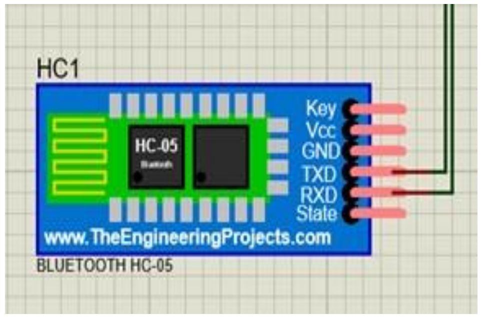
In this project arduino IDE is crucial because we are simulating the circuit in proteus. Arduino ide is generally used to upload the code to the arduino hardware but in our case we are using an arduino module inside proteus.

We type the code to the editor compile and in this case we generate a .hex file. This .hex file is a temporary file which is stored. We give the path of the .hex file to the virtual arduino present inside the proteus software.

Bluetooth Module

HC-05 Bluetooth Module is an easy to use Bluetooth SPP (Serial Port Protocol) module, designed for transparent wireless serial connection setup. Its communication is via serial communication which makes an easy way to interface with controller or PC.

- Bluetooth module acts as an interface between the app and arduino.
- It receives the signal from the phone and transmits it into arduino.
- It enables the wireless control of our circuit by accessing the Bluetooth of the laptop.
- Inside the circuit the transmission pin of the Bluetooth module is connected to pin no ‘0’ and the receiver pin is connected to pin no ‘1’ of the Arduino.



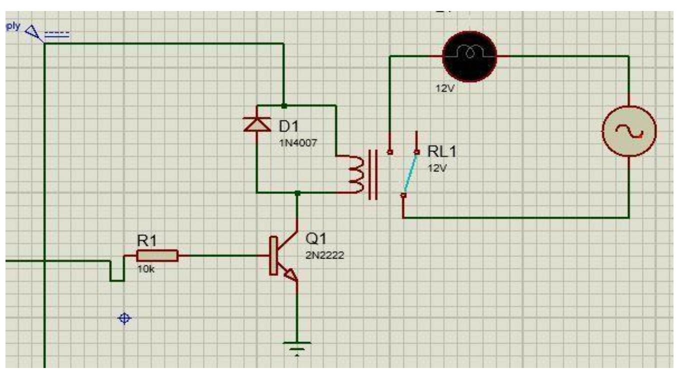
Relay switching circuit

Relay switching circuit acts as a virtual switch to our circuit. It contains relay module a npn-transistor, diode and a dc voltage source.

One end of the relay switching circuit is connected to arduino, when the arduino give high output to the switching circuit the transistor switches on as a result max current flows through the relay coil which causes it to switch on and the required power is delivered to home appliances (in this case LED and dc-motor).

When the arduino gives low output the transistor moves into the cutoff region which moves the relay switch back to initial position and the diode is used to discharge the coil. In our circuit we operate two relay switching circuit which are connected to pin '5' and '6' of the arduino.

- Relay switching circuit acts as a virtual switch for our appliances.
- It is connected to arduino which controls the action of the circuit.
- We use two relay switching circuits in our project to control the appliances



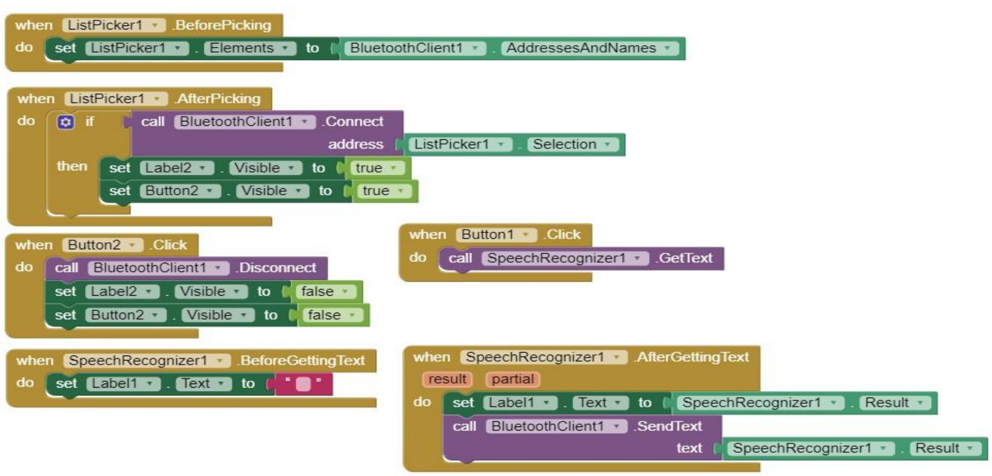
Mobile application

Mobile application is the interface between the user and the circuit. In our project we designed an android application using MIT app inventor. The mobile application contains a Bluetooth client and voice to text converter.

This application is connected to our circuit inside proteus virtually by using the COM-port of our laptop. The app is designed to convert our voice into text and send the text to our Bluetooth module wirelessly.

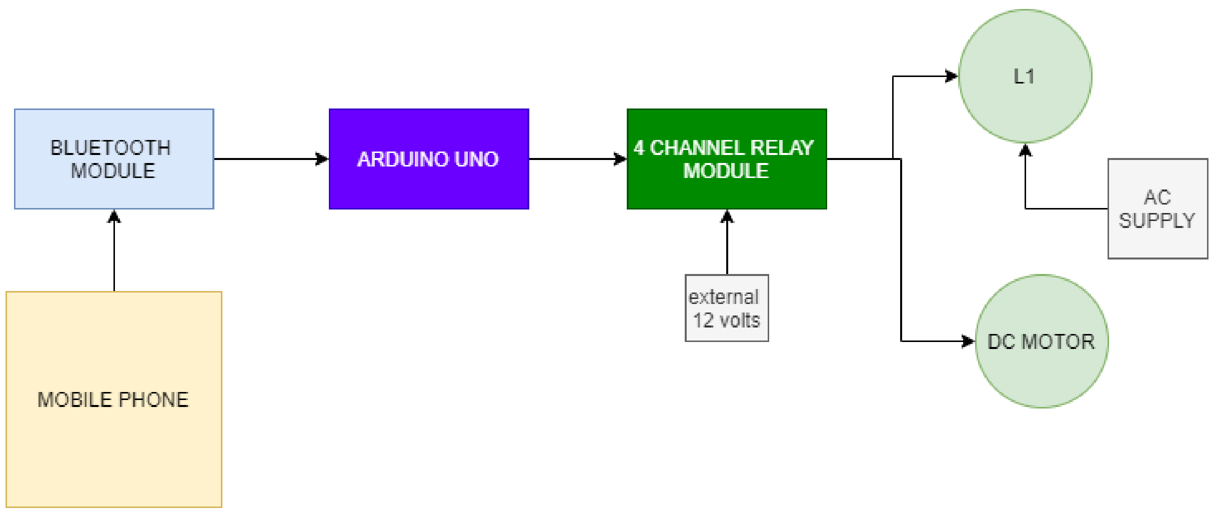
- Mobile application is where we give voice commands to control our project.
- The app reads the voice and converts it into text.
- The text is sent to Bluetooth module which is forwarded to arduino.

User interface

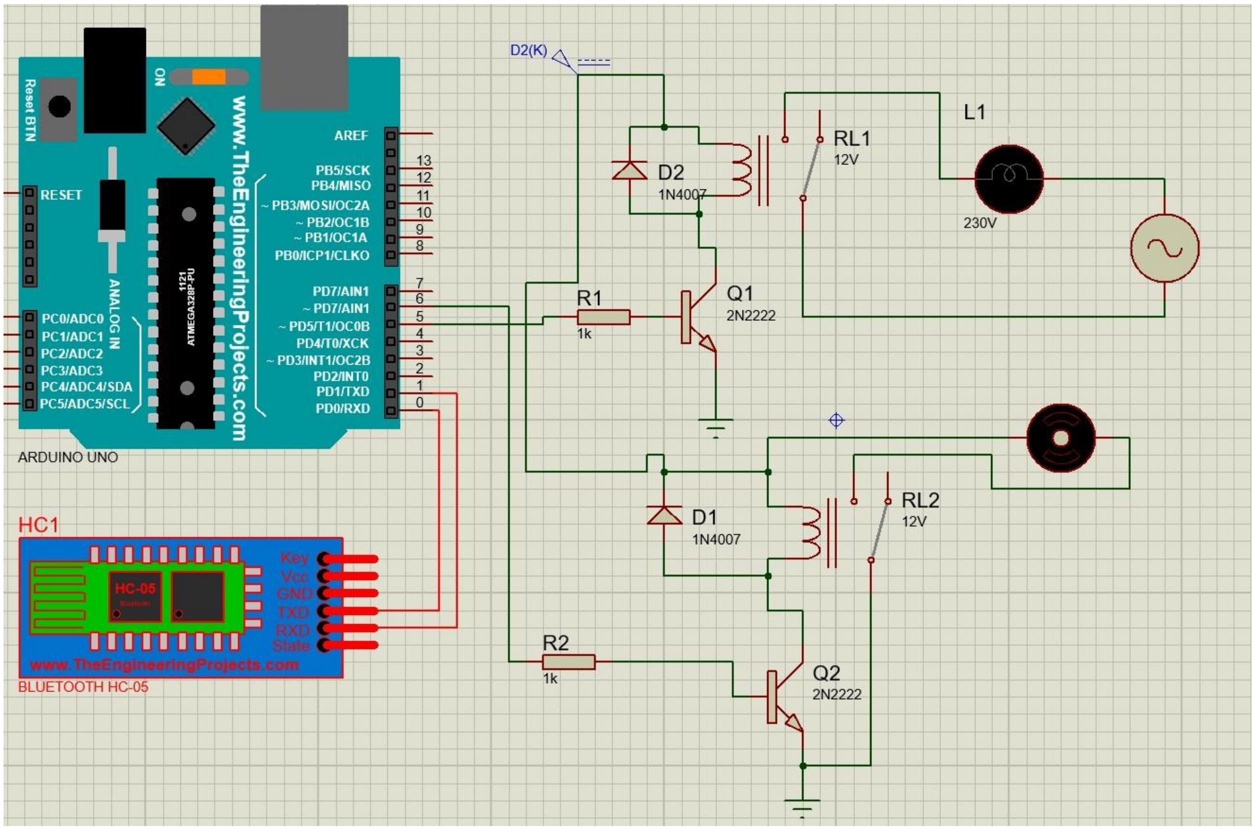


Backend design

BLOCK DIAGRAM:



CIRCUIT DIAGRAM(In Proteus):

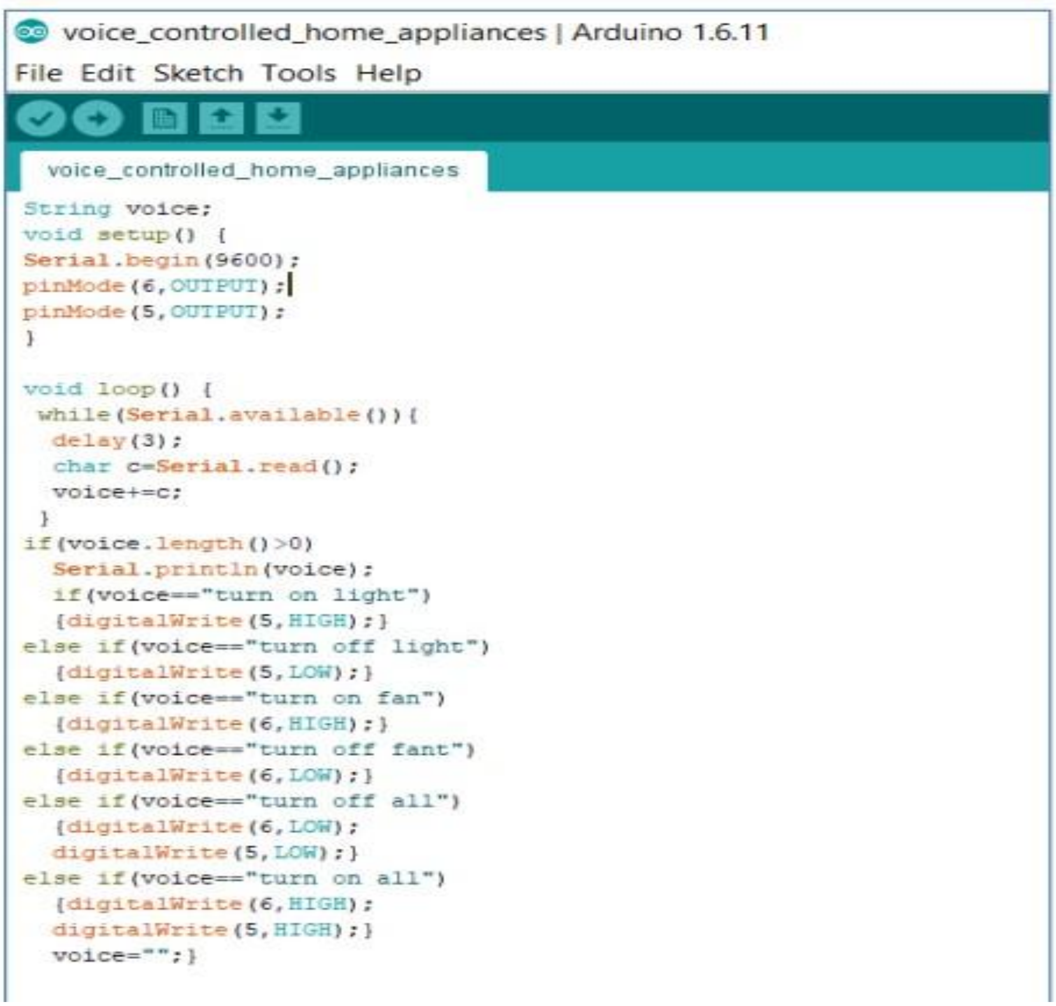


METHODOLOGY:

- Firstly, connecting all the components in the proteus software
- We need to upload the code into the Arduino uno with the help of Arduino software.
- We have to create a app using MIT app inventer from where we can connect the Bluetooth module in proteus with microphone in android.

- We should connect the app to Bluetooth module which contains the option for Bluetooth connection and microphone.
- If we want to turn on an then we should give the command “turn on fan”, for light “turn on light” and similarly for turn off through microphone.

ARDUINO CODE USED:


voice_controlled_home_appliances | Arduino 1.6.11

File Edit Sketch Tools Help

voice_controlled_home_appliances

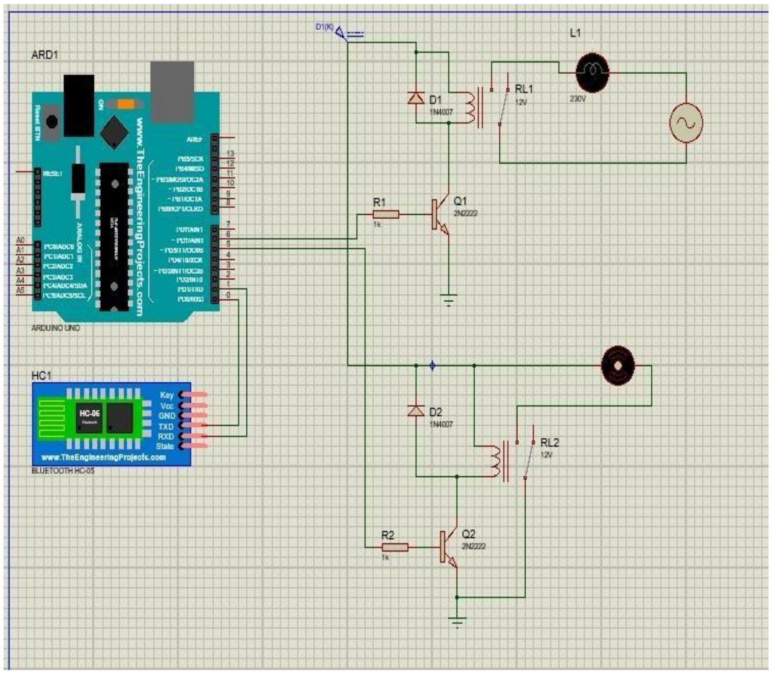
```
String voice;
void setup() {
Serial.begin(9600);
pinMode(6,OUTPUT);
pinMode(5,OUTPUT);
}

void loop() {
 while(Serial.available()){
  delay(3);
  char c=Serial.read();
  voice+=c;
 }
if(voice.length()>0)
  Serial.println(voice);
  if(voice=="turn on light")
  {digitalWrite(5,HIGH);}
else if(voice=="turn off light")
  {digitalWrite(5,LOW);}
else if(voice=="turn on fan")
  {digitalWrite(6,HIGH);}
else if(voice=="turn off fant")
  {digitalWrite(6,LOW);}
else if(voice=="turn off all")
  {digitalWrite(6,LOW);
  digitalWrite(5,LOW);}
else if(voice=="turn on all")
  {digitalWrite(6,HIGH);
  digitalWrite(5,HIGH);}
  voice="";}
```

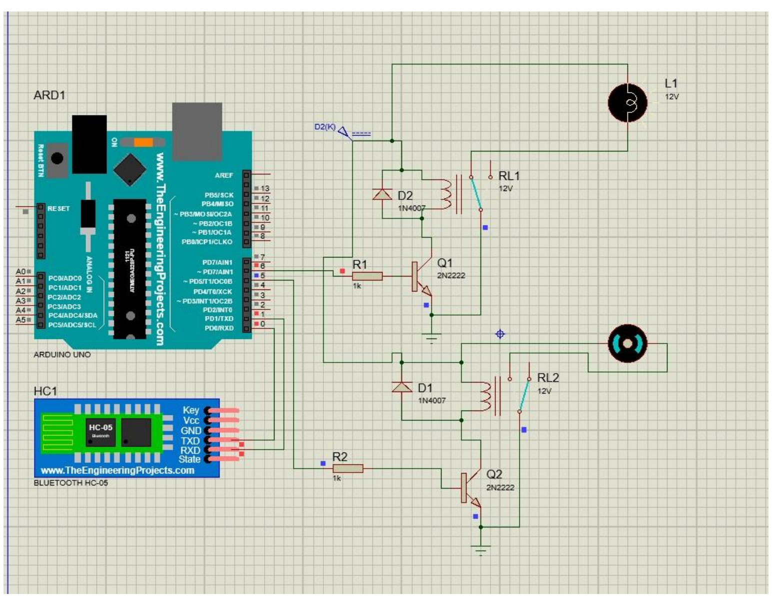
IOT APP LAYOUT:

RESULTS:

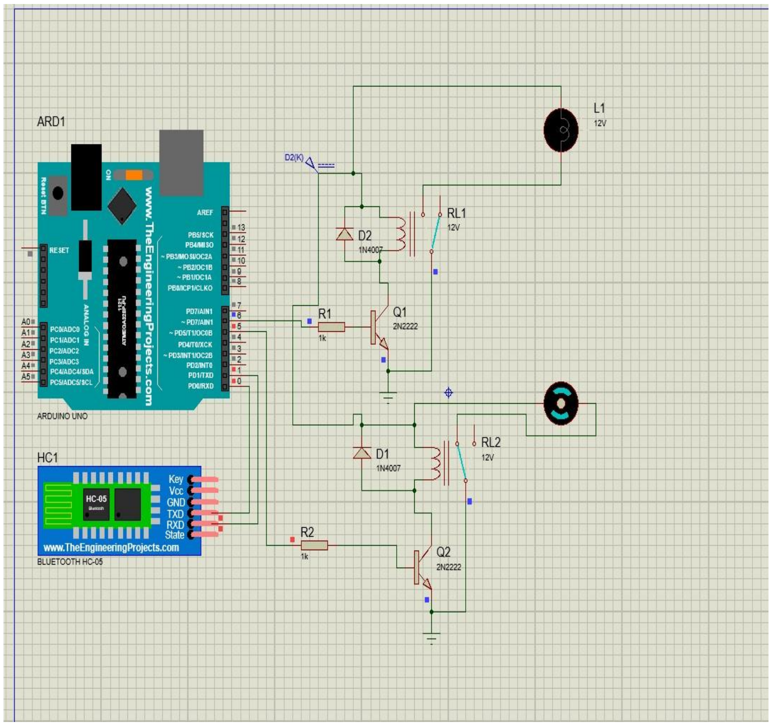
WHEN BOTH ARE OFF:



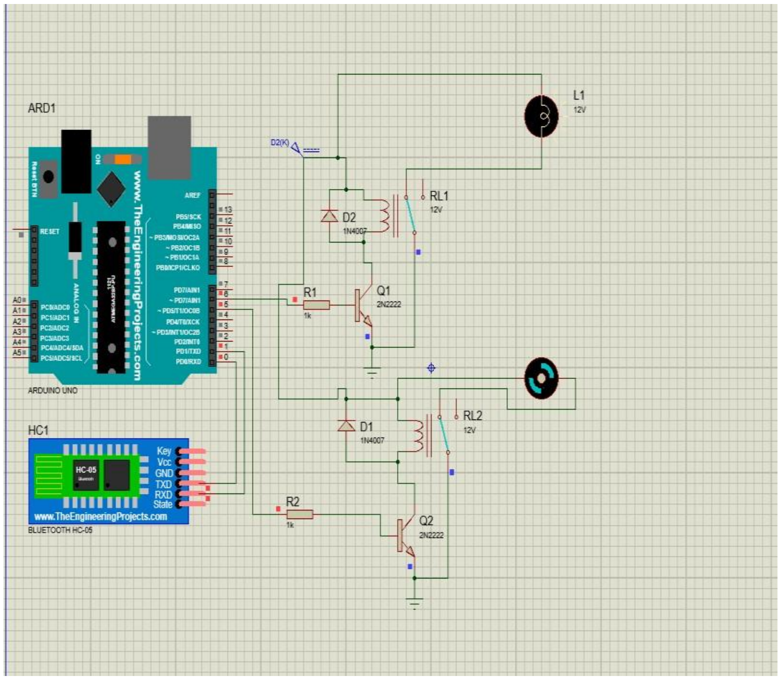
WHEN LIGHT IS ON:



WHEN FAN IS ON:



WHEN BOTH ARE ON :



CONCLUSION:

- Through this project we learnt to simulate hardware components on a software.
- We also learnt about the importance of home automation in our daily lives.
- We got to know about the working of the Bluetooth module and through this how we can control almost any appliance in our homes.

FUTURE WORK:

We can enhance this project by using real hardware components and implement the circuit in any of the appliance in our homes and can be enhanced to control the speed of the fan etc.

We can also use a wifi module for instead of a Bluetooth module for better productivity and results for a particular home appliance and also GSM modem can be used to achieve device controllling by sending SMS using GSM modem.

We can implement various sensors for more ease of usage for home applications.

REFERENCES:

[1] M. Asadullah and K. Ullah, "Smart home automation system using Bluetooth technology," 2017 International Conference on Innovations in Electrical Engineering and Computational Technologies (ICIEECT), Karachi, 2017, pp. 1-6, doi: 10.1109/ICIEECT.2017.7916544.

[2] K. Mandula, R. Parupalli, C. A. S. Murty, E. Magesh and R. Lunagariya, "Mobile based home automation using Internet of Things(IoT)," 2015 International Conference on Control, Instrumentation, Communication and Computational Technologies (ICCICCT), Kumaracoil, 2015, pp. 340-343, doi: 10.1109/ICCICCT.2015.7475301.